Comparing models with ANOVA

Each part has two scenarios with the same tasks. We'll work through scenario **A** together as a class. Then you'll work through scenario **B** in your group.

Every question in this handout compares a **simpler** model to a more **complex** model. Refer back to this box as you go.

$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \;/\; (p_{complex} - p_{simple})}{\text{RSS}_{complex} \;/\; (n - p_{complex})}$$

- RSS: residual sum of squares, $\sum(y_i - \hat{y}_i)^2$. The variation the model leaves unexplained.
- p: number of coefficients the model estimates, including the intercept
- n: number of observations
- $df = n - p$: degrees of freedom left over after fitting the model
- **Numerator**: improvement in fit per df spent. **Denominator**: leftover noise per df remaining.

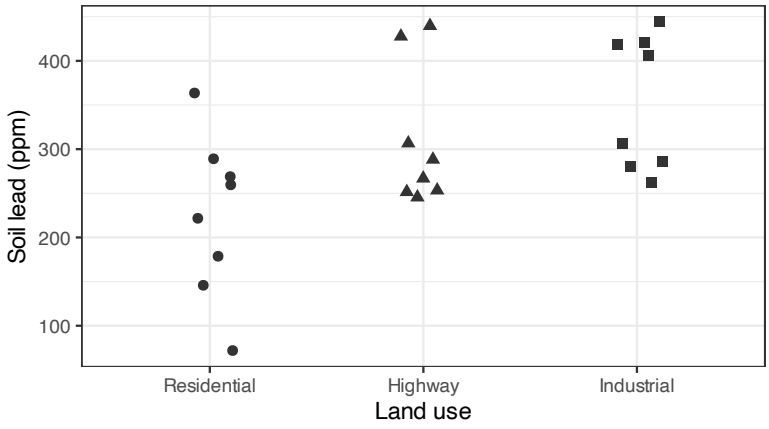
Part 1: Three levels

For each scenario:

1. **Data.** Look at the data in the figure.
2. **Dummy variables.** Fill in the dummy variable table.
3. **Full model.** Write the equation for the full model. What does each coefficient mean?
4. **Draw the means.** Use the coefficients to draw the intercept-only model and each category's mean on the figure.
5. **Degrees of freedom and F.** Use n and the RSS values to fill in p and df for each model. Then calculate F.
6. **Interpretation.** State H_0 and H_A. Then interpret the p-value for your F in a sentence.

1A. Soil lead (together)

Lead from old gasoline and industrial emissions settles into soil, where children can be exposed. A county health department sampled soil **lead** (ppm) in 8 yards near each of three land uses (**land_use**): residential streets, a highway, and an industrial site.



Dummy variables

Land use	Highway	Industrial
Residential	_____	_____
Highway	_____	_____
Industrial	_____	_____

Full model

Draw the means

```
coef(lm(lead ~ 1, yards))
```

```
(Intercept)
   295.9625
```

```
coef(lm(lead ~ land_use, yards))
```

```
       (Intercept)    land_useHighway land_useIndustrial
          224.9125            85.0625           128.0875
```

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`lead ~ 1`	_____	_____	210,538.5
`lead ~ land_use`	_____	_____	142,556.7

Interpretation

State H_0 and H_A.

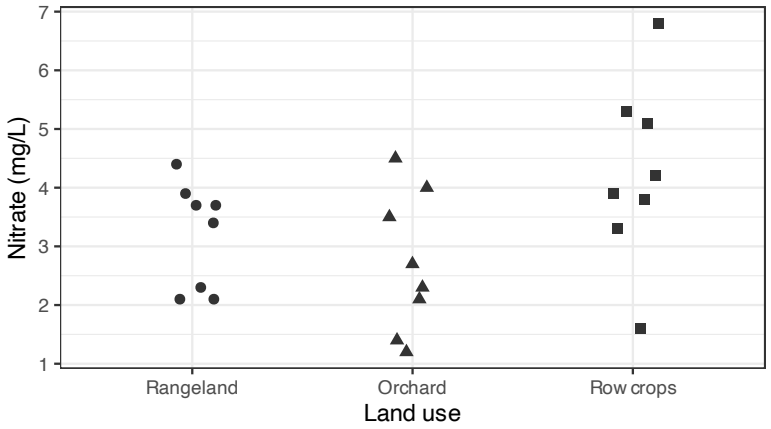
H_0:

H_A:

The p-value for this F is p = 0.017. Interpret it in a sentence.

1B. Groundwater nitrate (in groups)

Fertilizer applied to crops can leach into groundwater as nitrate, contaminating drinking water wells. A water agency measured **nitrate** (mg/L) in 8 wells under each of three land uses (**land_use**): rangeland, orchards, and row crops.



Dummy variables

Land use	Orchard	Row crops
Rangeland	_____	_____
Orchard	_____	_____
Row crops	_____	_____

Full model

Draw the means

```
coef(lm(nitrate ~ 1, wells))
```

```
(Intercept)
     3.3875
```

```
coef(lm(nitrate ~ land_use, wells))
```

```
      (Intercept)   land_useOrchard land_useRow crops
           3.2000           -0.4875            1.0500
```

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`nitrate ~ 1`	_____	_____	42.2
`nitrate ~ land_use`	_____	_____	32.3

Interpretation

State H_0 and H_A.

H_0:

H_A:

The p-value for this F is p = 0.061. Interpret it in a sentence.

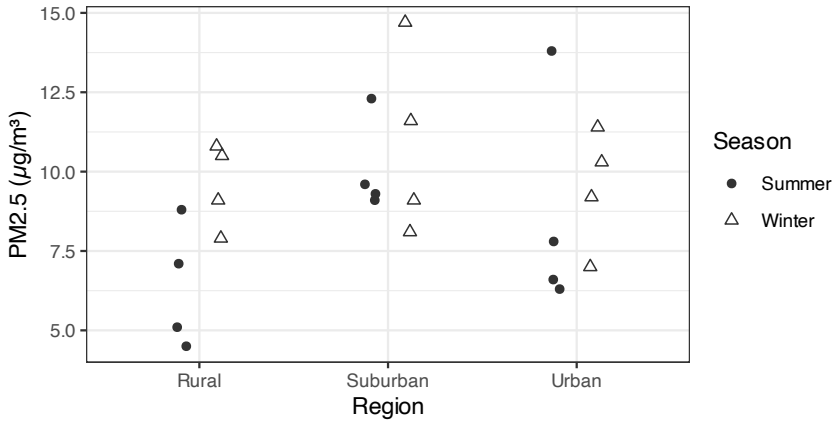
Part 2: Two categorical predictors

Now there are two categorical predictors. The question is whether the second one explains enough variation to justify its cost *after* accounting for the first. For each scenario:

1. **Data.** Look at the data in the figure.
2. **Full model.** Write the equation for the model with both predictors. What does each coefficient mean?
3. **Draw the models.** Use the coefficients to draw the simpler model's prediction for each level of the first predictor, and the model with both predictors' prediction for each combination of the two. Draw each prediction as a horizontal line segment, like in Part 1.
4. **Degrees of freedom and F.** Use n and the RSS values to fill in p and df for each model. Then calculate F for adding the second predictor.
5. **Interpretation.** State H_0 and H_A. Then interpret the p-value for your F in a sentence.

2A. Air quality (together)

Fine particulate matter (PM2.5) damages lungs and hearts. In winter, wood smoke and temperature inversions can trap it near the ground. An air district measured PM2.5 (**pm**, μg/m³) on 4 days for each combination of **region** (rural, suburban, or urban) and **season** (summer or winter). Does season matter after accounting for region?



Full model

Draw the models

```
coef(lm(pm ~ region, air))
```

```
   (Intercept) regionSuburban    regionUrban 
         7.975          2.500          1.075 
```

```
coef(lm(pm ~ region + season, air))
```

```
   (Intercept) regionSuburban    regionUrban   seasonWinter 
      7.166667       2.500000       1.075000       1.616667 
```

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`pm ~ region`	_____	_____	120.4
`pm ~ region + season`	_____	_____	104.7

Interpretation

State H_0 and H_A.

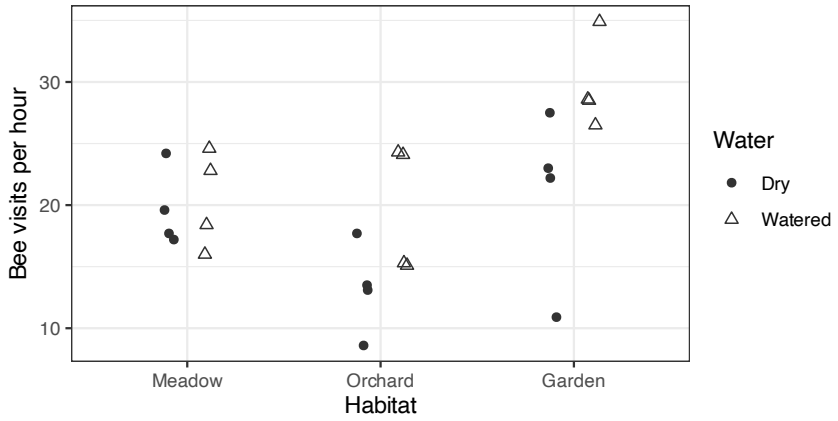
H_0:

H_A:

The p-value for this F is p = 0.099. Interpret it in a sentence.

2B. Bees and watering (in groups)

Watered flowers may produce more nectar, attracting more bees. A pollinator team counted bee **visits** per hour to 4 flower patches for each combination of **habitat** (meadow, orchard, or garden) and **water** (dry or watered). Does watering matter after accounting for habitat?



Full model

Draw the models

```
coef(lm(visits ~ habitat, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden
       20.0625        -3.6000         5.2000
```

```
coef(lm(visits ~ habitat + water, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden   waterWatered
        17.400         -3.600          5.200          5.325
```

Degrees of freedom and F (n = 24)

Model	p	df	RSS
`visits ~ habitat`	_____	_____	626.7
`visits ~ habitat + water`	_____	_____	456.5

Interpretation

State H_0 and H_A.

H_0:

H_A:

The p-value for this F is p = 0.013. Interpret it in a sentence.

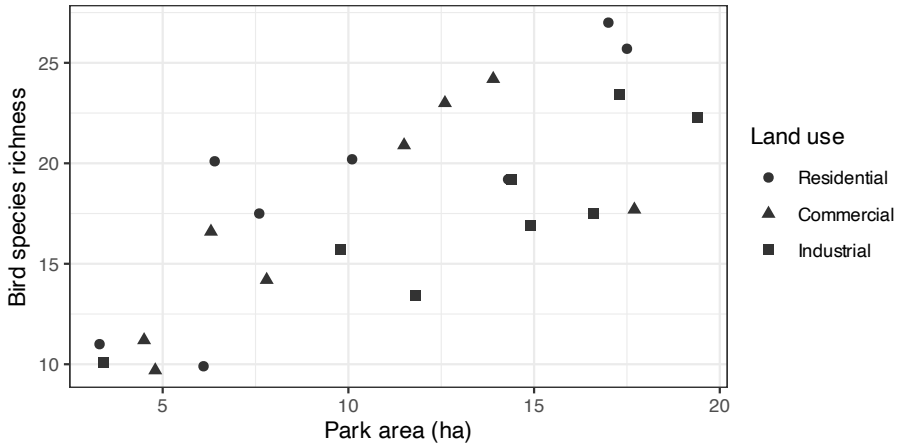
Part 3: Three levels plus a continuous predictor

Now the second predictor is continuous instead of categorical. The question is whether the categorical predictor explains enough variation to justify its cost *after* accounting for the continuous predictor. For each scenario:

1. **Full model.** Write the model equation with both predictors. What does each coefficient mean?
2. **Draw the models.** Use the coefficients to draw the model with only the continuous predictor (one line) and the model with both predictors (three parallel lines) on the figure.
3. **Degrees of freedom.** Fill in p and df for each model.
4. **F by hand.** Use the RSS values to calculate F for adding the categorical predictor.
5. **Check.** Compare your F to the `anova()` output.
6. **Conclusion.** State your conclusion in a sentence, using the p-value.

3A. Birds in city parks (together)

City parks can be refuges for birds, and bigger parks usually hold more species. Does what surrounds a park matter too? A city surveyed bird species **richness** in 8 parks in each of three surrounding land uses (**land_use**): residential, commercial, and industrial. It also recorded each park's **area** (ha).



Full model

Draw the models

```
coef(lm(richness ~ area, parks))
```

```
(Intercept)        area
  9.2357591   0.7618654
```

```
coef(lm(richness ~ area + land_use, parks))
```

```
       (Intercept)              area land_useCommercial land_useIndustrial
         9.9183439         0.8657746         -1.2911902         -4.2505121
```

Degrees of freedom (n = 24)

Model	p	df	RSS
`richness ~ area`	_____	_____	257.9
`richness ~ area + land_use`	_____	_____	188.4

F by hand

Check

```
anova(lm(richness ~ area, parks), lm(richness ~ area + land_use, parks))
```

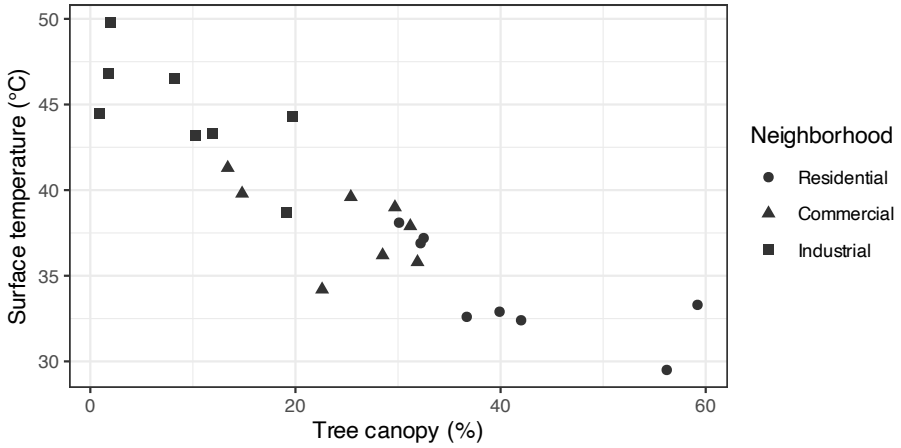
```
Analysis of Variance Table

Model 1: richness ~ area
Model 2: richness ~ area + land_use
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     22 257.92
2     20 188.44  2    69.475 3.6867 0.04335 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion

3B. Urban heat (in groups)

Pavement and rooftops soak up summer sun, while trees shade the ground. A city measured summer surface temperature (**temp**, °C) at 8 sites in each of three **neighborhood** types: residential, commercial, and industrial. It also recorded the tree **canopy** cover (%) around each site. Does neighborhood type matter *after* accounting for canopy?



Full model

Draw the models

```
coef(lm(temp ~ canopy, sites))
```

```
(Intercept)      canopy
 46.3901866  -0.2991744
```

```
coef(lm(temp ~ canopy + neighborhood, sites))
```

```
          (Intercept)                canopy neighborhoodCommercial neighborhoodIndustrial
           43.2780196            -0.2230053              0.2024248              3.4194922
```

Degrees of freedom (n = 24)

Model	p	df	RSS
`temp ~ canopy`	_____	_____	122.1
`temp ~ canopy + neighborhood`	_____	_____	96.1

F by hand

Check

```
anova(lm(temp ~ canopy, sites), lm(temp ~ canopy + neighborhood, sites))
```

```
Analysis of Variance Table

Model 1: temp ~ canopy
Model 2: temp ~ canopy + neighborhood
  Res.Df     RSS Df Sum of Sq     F  Pr(>F)
1     22 122.065
2     20  96.076  2    25.988 2.705 0.09126 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion